Hiba Malik

Software Engineer | Full Stack & Systems

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PROFILE

Software Engineer with 4 years of experience building full-stack and backend systems across **Web2** and **Web3**. Experienced in designing APIs, database-driven applications, real-time systems, cloud infrastructure, and distributed applications, with a strong focus on backend engineering and system architecture. I have independently owned technical domains end-to-end, from understanding requirements and researching solutions to making architectural decisions, implementing systems, and integrating them into production applications. My experience spans **AI-enabled** applications, **blockchain** infrastructure, and **complex software products**, and I am currently expanding my expertise in AI engineering and LLM-powered systems.

EDUCATION

COMSATS University, Bachelor's in Software Engineering 2018 – 2022 | Pakistan
Final year project: Health record management using Blockchain
Achievement: Silver Medalist

PROFESSIONAL EXPERIENCE

DroppGroup, Blockchain Developer/Full Stack May 2025 – May 2026 | Dubai, UAE

- Led the **launch of the D-GN** token on PumpFund, achieving **30,000+ transfers** within weeks post-launch.
- Designed & deployed **Solana vesting contracts**, enabling scheduled release of ~600M DGN tokens from vaults.
- Developed a vesting dashboard in **Next.js** integrating vesting contracts and Fireblocks authentication to display real-time token balances, release schedules, and secure token claim execution with live contract state synchronization.
- Provisioned and integrated **Fireblocks vaults** to securely lock large token volumes prior to vesting.
- Developed a **dual-chain Substrate-based blockchain** with custom consensus and integrated a custom chain explorer using **Nest** and **Next.js**.

ArgonTech, Blockchain/Backend Developer July 2022 – April 2025 | Lahore, Pakistan

- Designed and implemented **off-chain signing** on backend, enhancing transaction security and reducing on-chain gas costs by 30%.
- Worked as DevOps for deploying blockchain and backend servers on **AWS** & **Google Cloud**.
- Engineered **gas-free** blockchain transactions using Thirdweb's account abstraction, enabling seamless interaction for both crypto and non-crypto users, increasing user adoption by 60%.
- Built a robust cron job system to monitor and retry failed blockchain transactions, improving **marketplace reliability** by 95%.
- Received 97% score in **Hacken Audit** for smart contracts.

LANGUAGES

English (Native) Arabic (Conversational)

SKILLS

Languages — Python | TypeScript | JavaScript | Go | Rust | Solidity

Backend — NodeJS | NestJS | FastAPI | REST | GraphQL | Websockets

Databases — PostgreSQL | MongoDB

Cloud / Infrastructure — AWS | Google Cloud | Docker | Linux

Frontend — ReactJS | NextJS | Web3 Wallets

Distributed Systems / Blockchain — Ethereum | Solana | Substrate | Hyperledger

AI — LLM APIs | Generative AI | AI Agents | RAG

PROJECTS

droppRWA, Real-World Asset Tokenization Platform

- Developed **full-stack** blockchain infrastructure and backend services for a **dual-layer** Substrate ecosystem powering tokenized real-world assets.
- Built APIs and smart contract integrations supporting autonomous AI-driven asset settlement.
- Contributed to the **droppOne** platform — an agentic AI system integrating multimodal AI models with smart contracts.
- Worked on **agentic AI workflows** enabling autonomous execution, settlement, and governance of tokenized assets on-chain.
- Contributed to Saudi Arabia's **first-ever tokenized property title deed transfer** (February 2026)

Vesting Dashboard, Solana Vesting Platform

- Architected & developed a vesting management platform for crypto token distribution program.
- Built backend blockchain indexers and contract readers to aggregate and cache vesting state for multiple concurrent vesting programs.
- Implemented secure Fireblocks wallet integration with role/account validation.
- Optimized frontend state management to selectively refresh updated vesting components after claim execution without full-page reloads with resusable React Components.
- Implemented real-time transaction monitoring, error handling, and blockchain confirmation.
- **Tools & Technologies**: Next.js, Node.js, MongoDB, Solana Web3.js, Fireblocks SDK, TailwindCSS

Katana Inu, In-game NFT Marketplace

- Engineered a diverse Solidity blockchain suite, including staking, and marketplace functionalities.
- Implemented factory patterns for efficient ERC721 contract deployment.
- Enhanced gas efficiency & security using EIP712, benefiting NFT listing with improved user flexibility.
- Designed a real-time WebSocket communication layer between Electron and React applications for persistent wallet sessions, live transaction synchronization, and low-latency MetaMask event handling.
- Designed a robust back-end system for NFT metadata storage and affiliate management allowing users to earn commissions by promoting NFTs.
- Created customizable referral links and tracking for affiliates.
- **Tools & Technologies:** Solidity, EthersJs, Web3, Websockets, NodeJs, MongoDb

Galileo Protocol, Physical NFT Marketplace

- Developed a centralized database using PostgreSQL to track NFT metadata and transaction status.
- Created cron jobs to monitor blockchain transactions and re-initiate failed transactions for consistency.
- Leveraged **SubGraph's** GraphQL to display comprehensive transaction history.
- Integrated MetaMask SDK and Magic Wallet SDK into a ReactJS frontend for smooth user interaction.
- **Tools & Technologies:** PostgreSQL, EthersJs, RESTful Apis, GraphQL, ReactJS, Node.js